

135.71% Above Pre-Pandemic Levels: A Data-Led eSIM Market Expansion Strategy

Executive Summary

The eSIM market is shifting from volume-driven expansion to value-driven targeting efficiency. The data shows a structural disconnect between arrival volumes and revenue generation.

Key findings:

- Revenue growth is increasingly decoupled from arrival growth, with some markets exceeding +130% expenditure growth vs 2019.
- Tourism demand is highly concentrated, with the top 20% of countries generating 76.3% of global arrivals.
- High-value markets are emerging outside traditional premium destinations, particularly in recovery and emerging economies.

Key recommendation:

Reallocate acquisition spend toward high-value premium markets where expenditure growth significantly outpaces arrival growth (notably El Salvador, Saudi Arabia, India, Mexico).

Business Problem

GlobalComm is currently allocating marketing spend based on raw arrival volumes, rather than revenue potential. This overlooks high-value premium market where expenditure per travel is growing 2X faster than arrivals. Without a data-driven prioritization framework, the company risks over-investing in low-margin saturated markets while missing high-LTV (Lifetime Value) opportunities in recovering regions.

Key Questions:

- Which destinations have the highest ratio of Expenditure to Arrivals, and how does this correlate with Accommodation Capacity?
- Which regions have surpassed 2019 arrival levels the fastest (2022–2024), and is their tourism spending growing faster than their arrival volume?
- In markets with high 'Inbound arrivals by mode of transport - Air', what is the average length of stay?
- How does the 'Main Purpose of Trip' (Business vs. Leisure) correlate with the average tourism expenditure in top-performing regions?
- What is the variance in monthly/quarterly arrival data across our top 10 Primary Markets?
- Based on findings in above analysis, what 3 key actions should GlobalComm implement, and what impact will they have?

Data Cleaning & Processing:

- Filtered thousands of indicator rows to focus exclusively on Total Overnight Visitors (Tourists) and Total Travel Expenditure. This prevents double-counting of same-day visitors or non-travel financial flows.
- Merged capacity data (hotels/rooms) as a proxy for the stability of the destination's tourism infrastructure.

- Created the `exp_per_arrival` metric. In the context of "GlobalComm," this represents the purchasing power of the average traveler in that region, a critical indicator for eSIM product tiering (e.g., selling "Unlimited" vs. "Lite" plans).
- Focused on the 2020–2024 window to capture the recovery phase, which is more relevant for current market forecasting than pre-pandemic historical data.
- Dropped records with incomplete arrivals or expenditure values to ensure any derived scores are statistically sound.

Data Approach & Methodology

Data: UN Tourism Inbound Database (Hotels, Arrivals by Transport, Purpose of Trip, and Expenditure).

Tools: Python (Pandas, NumPy), Seaborn/Matplotlib for visualization; Coefficient of Variation for volatility; Pareto analysis for concentration; linear regression for growth slopes.

Data Cleaning & Processing:

- Filtered 10,000+ rows of raw metadata to isolate "Total Overnight Visitors" and "Total Inbound Expenditure," removing same-day excursionists and non-tourism financial flows.
- Focused on the 2020–2024 window to capture the recovery phase, which is more relevant for current market forecasting than pre-pandemic historical data.
- Markets with fragmented historical records were evaluated for structural consistency
- Expenditure per Arrival (EPA) was capped at the 95th percentile during visualization to ensure that global trends were not obscured by micro-state tax anomalies.
- 2019 serves as the pre-pandemic baseline, allowing for the calculation of a Recovery Slope (a measure of which markets have structurally expanded versus those that have rebounded).

Data Analysis:

The following key analytical frameworks were employed to answer the core business questions;

- **Yield-Friction Index:** A cross-correlation of Expenditure per Arrival vs. Accommodation Capacity. This identifies markets where traveler spending is high but local infrastructure is low - a critical indicator for digital-first connectivity solutions.
- **Stability Modeling (CV):** Applied the Coefficient of Variation across the 2020–2024 window. This allows for scoring each market's risk-adjusted growth, distinguishing between stable strongholds and volatile opportunities.
- **Pareto Distribution:** A cumulative concentration analysis on 2024 arrivals quantifies the degree of market saturation and justifies the strategic 20% budget reallocation.

Operational Impact:

This methodology transforms raw UN data into a prioritization framework, enabling GlobalComm to:

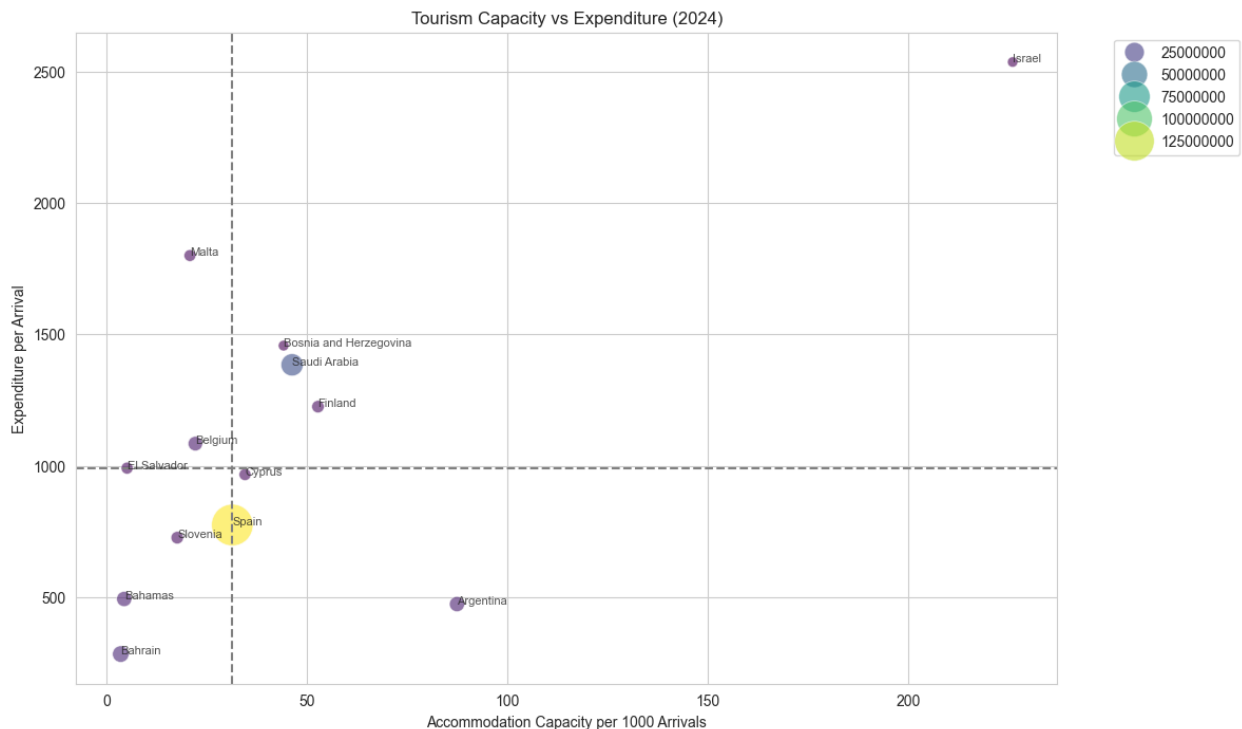
- Identify high-LTV corridors before they reach full saturation.
- Optimize acquisition bids based on Length of Stay rather than just click volume.
- Defend high-margin tiers in markets where spending is growing 2x faster than arrivals.

Key Findings

1. 76.3% of Global Arrivals are Concentrated in Just 20% of Markets.

These markets are highly saturated and increasingly price sensitive. Traditional marketing is fighting for the same low-margin travelers. The real opportunity lies in the "long tail" where the remaining 23.7% of travelers are spending at a significantly higher rate per arrival.

2. High-Yield Markets Face Infrastructure Friction



A positive correlation of 0.69 confirms that mature tourism markets typically scale infrastructure alongside traveler spend. However, outliers like El Salvador and Malta represent asymmetric opportunities: travelers spend at premium levels despite constrained accommodation supply. These markets signal high-value demand outpacing infrastructure, a connectivity gap eSIM can address without requiring physical asset investment.

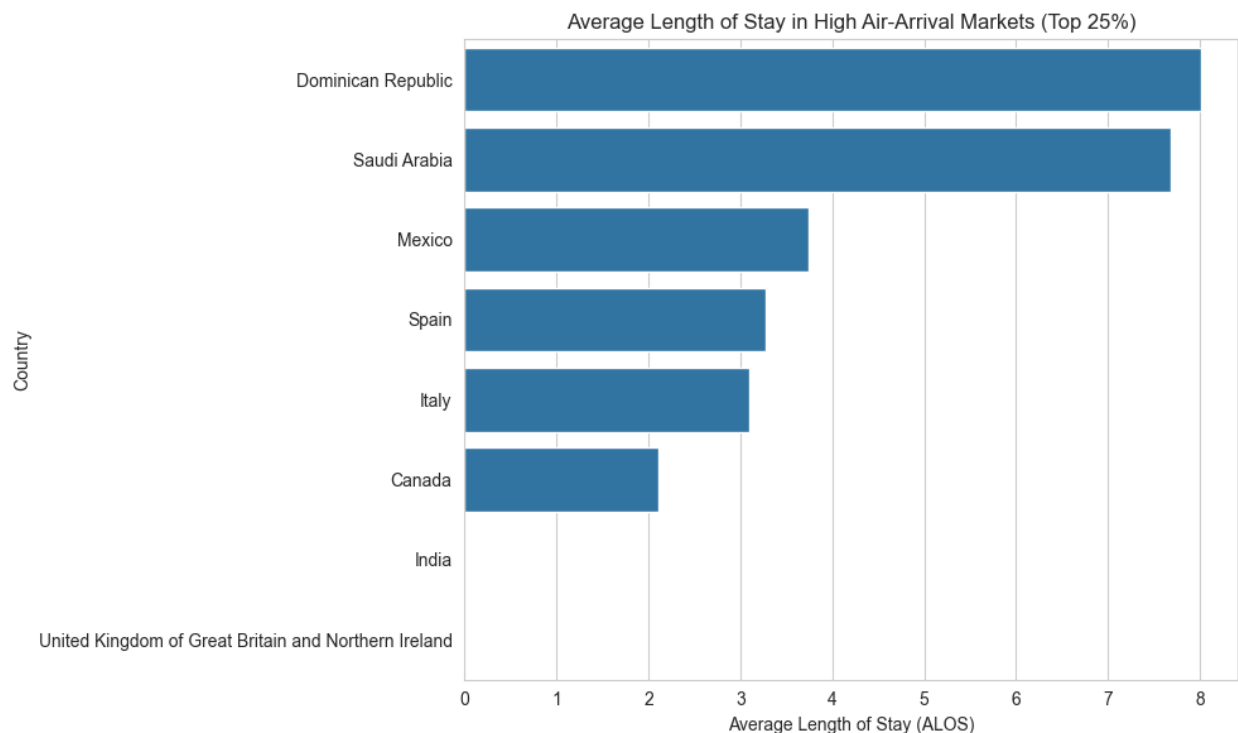
3. Markets Recovering Fastest Since 2019 Are Also Concentrating Revenue Growth in Select Destinations

Post-Pandemic Recovery Analysis (2024 vs 2019)

	Arrival_Growth	Expenditure_Growth	Value_Premium
country			
Saudi Arabia	64.99	133.45	68.46
Bahamas	54.92	33.57	-21.35
El Salvador	49.94	135.71	85.77
Bahrain	34.54	9.51	-25.03
Bosnia and Herzegovina	15.92	65.19	49.27
Mongolia	14.13	21.82	7.69
Slovenia	7.49	9.53	2.04
Belgium	3.15	-0.61	-3.76
Argentina	2.05	-4.76	-6.81
Trinidad and Tobago	0.93	57.06	56.13

The fastest markets to surpass 2019 arrival levels were Saudi Arabia(+64.99%), Bahamas (+54.92%), and El Salvador (+49.94%). However, recovery quality varied. El Salvador (+135.71%) and Saudi Arabia (+133.45%) significantly outperformed in tourism spending growth, while Bahamas (+33.57%) lagged arrival growth. These markets (where spending growth exceeds arrival growth) offer stronger monetization potential per acquired traveler.

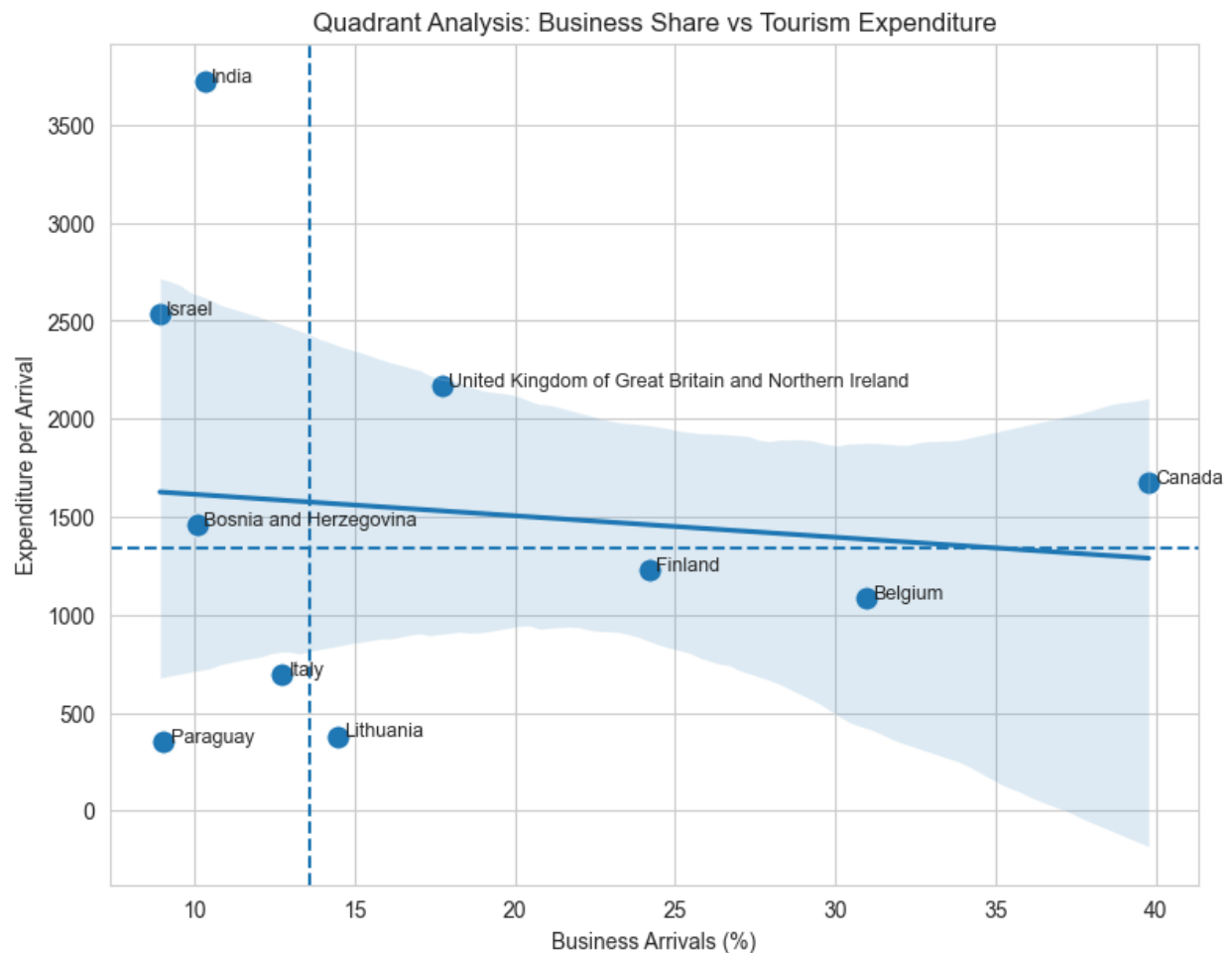
4. High Air-Arrival Markets Create Longer Usage Windows



Among top air-arrival markets, the longest average stays were recorded in Dominican Republic (8.0 days) and Saudi Arabia (7.7 days), well above markets such as Spain (3.3 days) and Italy (3.1 days). Overall, the

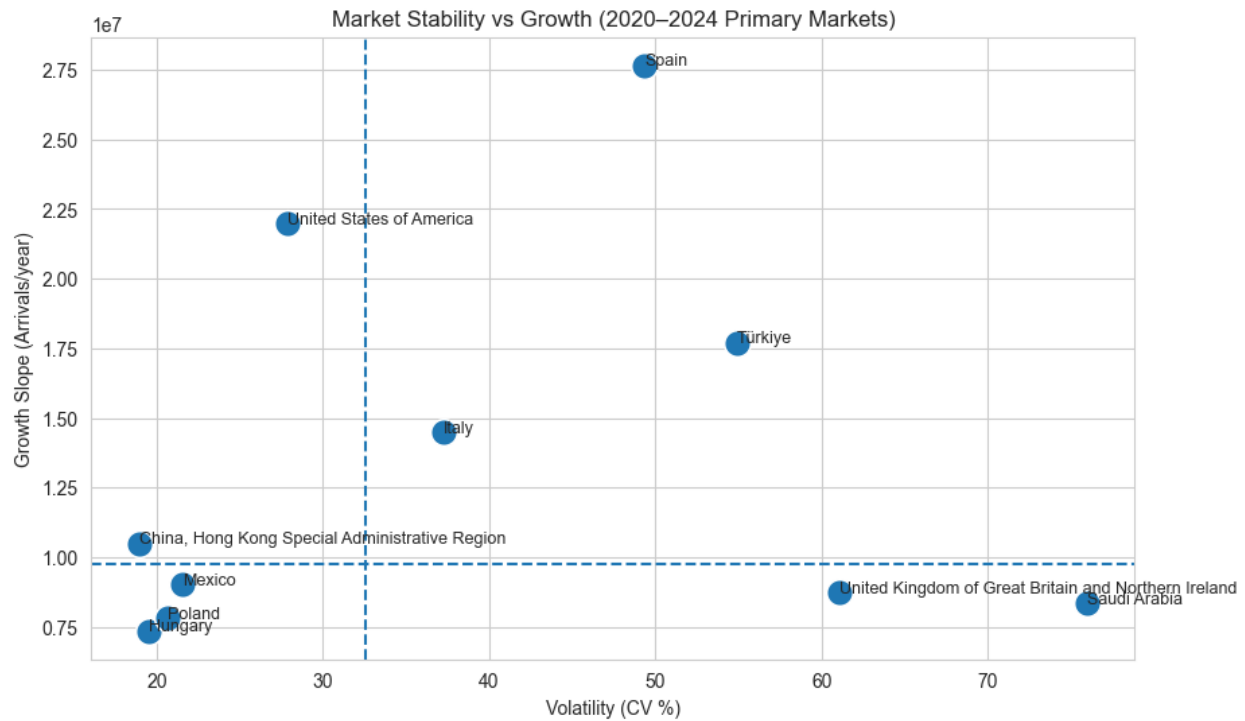
relationship between air arrival volume and stay duration was weak ($r = -0.18$), indicating that larger air traffic does not necessarily translate into longer stays.

5. Business Travel Markets Show Higher Spend per Traveler



Leading business-share markets included Canada (39.8%), Belgium (31.0%), and Finland (24.2%), all of which recorded above-average expenditure per arrival. The UK also combined a sizeable business segment with strong spend levels. The positive correlation ($r=0.23$) is modest but directional: markets with stronger corporate travel flows are better positioned for premium eSIM bundles, larger data plans, and higher conversion value per customer.

6. Primary Markets Differ Significantly in Stability and Recovery Pace



Across the top 10 primary markets (2020–2024), arrival volatility varied sharply, with Saudi Arabia (CV 76.0%), UK (61.1%), and Türkiye (54.9%) showing the highest fluctuations. More stable markets included Hong Kong SAR (18.9%), Hungary (19.5%), and Poland (20.6%). Recovery momentum was strongest in Türkiye, Saudi Arabia, and the United States, based on normalized growth trends.

Strategic Recommendations

- Shift budget from saturated volume markets to high-value recovery markets such as Saudi Arabia, El Salvador, and Bosnia & Herzegovina, where spend growth outpaces arrivals.
- Launch premium plans in business-heavy corridors (Canada, UK, Belgium) with higher data caps, roaming benefits, and priority support.
- Prioritize airport and arrival-channel acquisition in long-stay air markets to maximize top-up and repeat purchase potential.
- Use a balanced market portfolio strategy: combine high-growth volatile markets with stable demand markets to reduce CAC volatility.
- Replace arrival-based targeting with a market scoring model using spend per traveler, recovery growth, and stability metrics.